Active Directory Home Lab

Setup & Configuration Documentation

Platform: Oracle VirtualBox | OS: Windows Server 2022 + Windows 11

Overview

This lab simulates a basic corporate Active Directory environment using virtual machines. The goal is to get hands-on experience with Windows Server, domain setup, and user/policy management – all from a personal machine without needing real hardware.

Active Directory is one of the most widely used identity management systems in enterprise IT. Understanding how to set it up, configure it, and manage it is a core skill for roles in sysadmin, IT support, security, and cloud administration.

What You'll Need

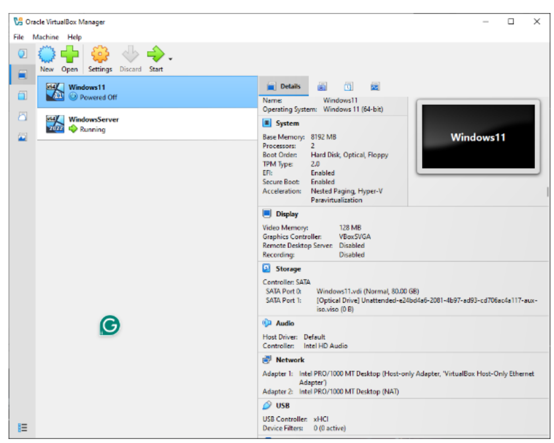
- **Oracle VirtualBox** – free virtualization platform
- **Windows 11 ISO** – as a client/workstation
- **Windows Server 2022 ISO** – act as a domain controller
- **Host machine specs:**
  - 16GB RAM or more (two VMs will share resources)
  - 130GB free disk space
  - A decent multi-core processor

Part 1: Virtual Machine Setup

Windows 11 (Client Machine)

This represents a regular workstation that will join the domain, similar to what an employee would use in an office.

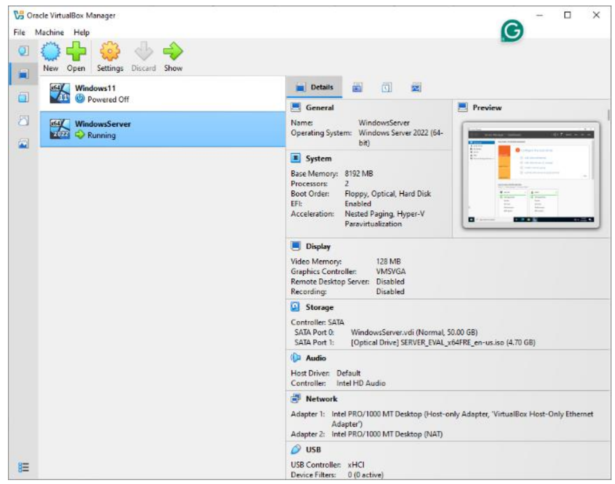
1. Download the Windows 11 ISO from Microsoft's official website.
2. In VirtualBox, create a new VM and attach the ISO as the boot disk
3. Set the following allocation:
   - Disk: 80GB (Default)
   - RAM: 8GB
   - CPU: 2 Cores
4. Boot and complete Windows installation as normal



Windows Server 2022 (Domain Controller)

This will be promoted to a Domain Controller (DC), meaning it manages all authentication and policies for the domain.

5. Download the Windows Server 2022 ISO from Microsoft's official website.
6. Create a new VM in VirtualBox and attach the ISO.
7. Set the following resource allocation:
   - Disk: 50 GB (Default)
   - RAM: 16 GB
   - CPU: 2 cores
8. During installation, select Windows Server 2022 Standard (Desktop Experience) – this gives you the GUI instead of command lines.
9. Complete the setup and create a strong Administrator password when prompted.

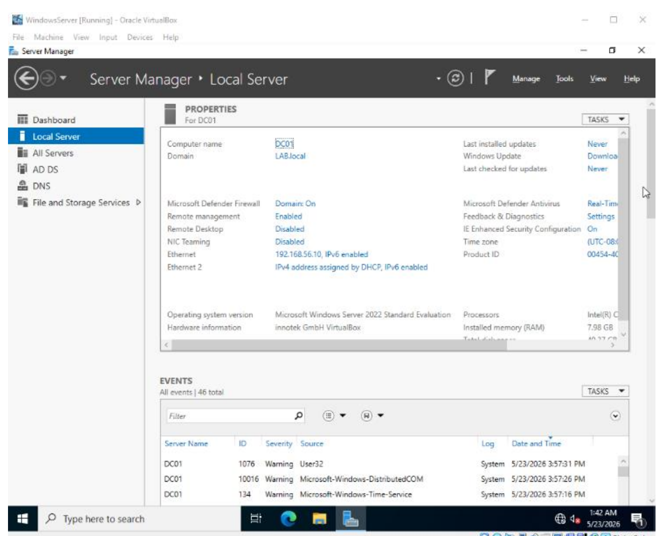


Part 2: Server Configuration

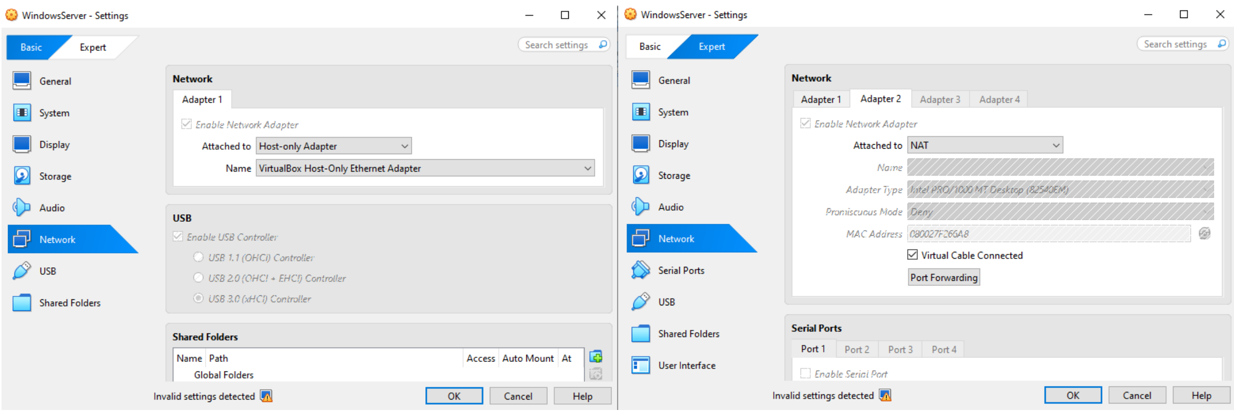
Step 1 – Rename the Server

Giving the server a recognizable name makes it easier to identify on the network, especially when you have multiple machines.

10. Open **Server Manager** and click on **Local Server** in the left panel.
11. Click the current computer name (shown next to the computer name field) to open System properties.
12. Click **Change** and enter your desired name (e.g., **DC01**).
13. Restart the server when prompted – the name won't take effect until you do.



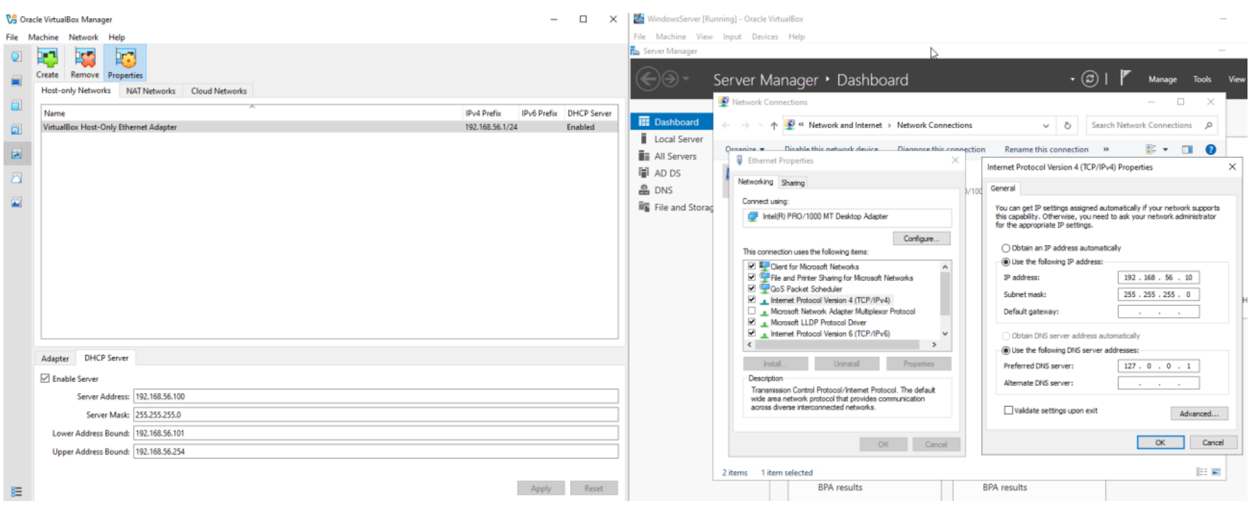
Step 2 – Configure the Network Adapter



Switching to a Host-Only adapter isolates the lab network from your real internet connection. This means any network issues or misconfiguration in the lab won't affect the actual network.

14. In VirtualBox, select the Windows Server VM and open **Settings > Network.**
15. Under Adapter 1, change **Attached to from NAT** to **Host-Only Adapter.**
16. For Adapter 2, leave it at **NAT.**
17. Click OK to save.

Step 3 – Assign a Static IP Address



A static IP ensures the server always has the same address on the network, which is required for clients to consistently find and connect to the Domain Controller.

18. First, check VirtualBox's DHCP range – go to **File > Tools > Network** to see the IP range in use under the **DHCP server**.
19. Inside the server VM, go to **View Network Connections.** Right-click the **Ethernet adapter > Properties.**
20. Select **Internet Protocol Version 4 (TCP/IPv4) > Properties.**
21. Select the **Use the following IP address** and enter:
    - **IP Address:** 192.168.56.10
    - **Subnet Mask:** 255. 255. 255. 0

- **Preferred DNS Server:** 192.168. 56. 10 (same as the server's IP – the DC will act as DNS)

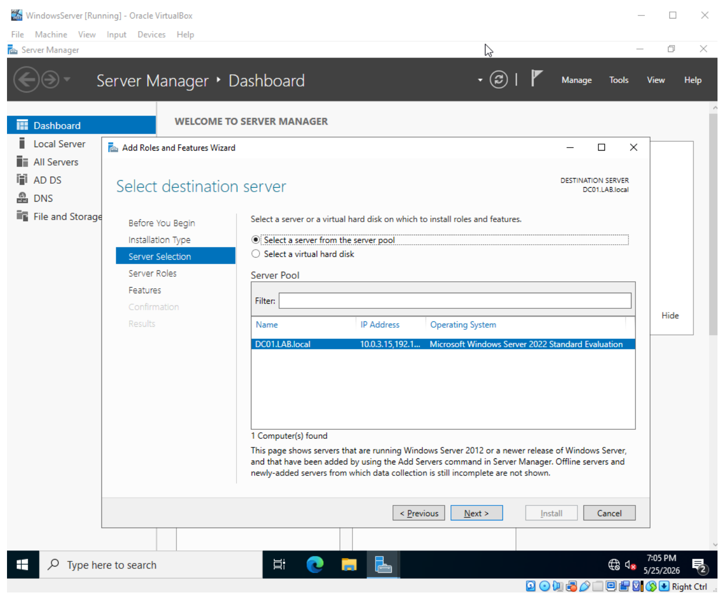
22. To verify, open Command Prompt and run **Ipcconfig**. Confirm the IP Address is set properly.

Part 3: Installing Active Directory

Step 1 – Install the AD DS Role

Active Directory Domain Services (AD DS) is the role/feature that enables the server to serve as a domain controller. Installing it is just the first step – you still need to promote the server afterward.

23. In Server Manager, click **Add Roles and Features.**
24. Click Next on the Before You Begin page.
25. Select **Role-based or feature-based Installation,** then click Next.
26. Keep the default server selected, then click Next.
27. Check the box for **Active Directory Domain Services.** When prompted, click **Add Features.**
28. Keep clicking Next, then click **Install.** Wait for it to complete.

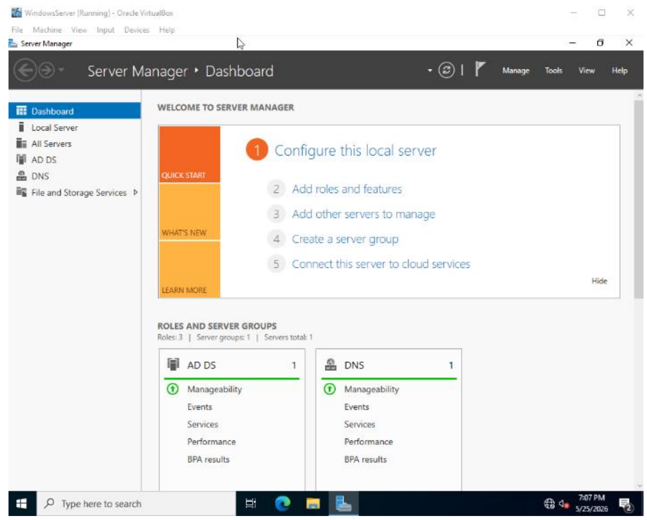


Step 2 – Promote to Domain Controller

After installing AD DS, the server needs to be "promoted" - this is the process of actually configuring it as a Domain Controller and creating the domain itself.

29. In Server Manager, click the yellow notification flag at the top and click **Promote this server to a domain controller.**
30. Select **Add a new forest** and set the Root domain name – e.g., **LAB.local**.
31. Click Next and set a strong **DSRM (Directory Services Restore Mode)** password – keep this somewhere safe.
32. The NetBIOS domain name (**LAB**) will be filled in automatically. Click Next.

33. Continue clicking Next through the remaining options and click **Install.**
34. The server will automatically restart to apply all changes. After reboot, log back in using the domain account (**LAB\Administrator**). AD tools will now be available under Administrative Tools.



Network Notes

One thing I wanted to point out about this lab setup is the configuration of the two VMs on their network adapters.

The Windows 11 and the Windows Server machines are both setup the same network adapters – a host-only adapter and a NAT adapter.

Windows 11 has internet access through the NAT adapter since network settings are automatically configured via DHCP, which assigns the IP address, subnet, and gateway with no manual configuration needed.

The Windows Server has a manually assigned static IP which point to itself. Since there is no default gateway configured, the server has no route to send traffic outside the local network, which prevents internet access from working as intended, even if NAT is configured.

In a realistic setting, domain controllers are kept off the internet anyway, and here it happened naturally as a result of IP configuration. But, if in need of internet access, it would have a default gateway pointing to the internal router, which then routes traffic out through the ISP (Internet Service Provider).

For DNS, admins configure DNS Forwarders on the Domain Controller – so internal domain traffic is handled by the Domain Controller itself, while external website lookups get forwarded to a public DNS like Google (8.8.8.8) or Cloudflare (1.1.1.1). In this lab, it was not configured, which is why there is no internet access, but for the Domain Controller's purpose, it really does not need it.